

Assignment #01

IT University of Copenhagen (ITU)
Data Mining and Machine Learning KSD (DAMML)
(Autumn 2026)

Deadline: 25 October 2026 at 23:59

Introduction For this mandatory assignment, your group will complete two tasks: (1) implement a set of mandatory exercises from the weekly exercise sessions (the ones with the tag (**Assignment #01**)), and (2) a small-scale data mining and machine learning report. In the second task, you will work with a dataset similar to the one created during Lecture #02 based on the questionnaire you all filled out. You must use this data to answer two or more questions you define. Then, you will implement and analyze the necessary code to answer those questions you have selected.

Purpose This assignment aims to give you a taste of what a data mining and machine learning project looks like. You will work with a dataset, implement the necessary code to analyze it and write a project report to present your findings. This assignment will also help you understand the importance of defining a straightforward question **before** starting a data mining and machine learning project. On the other hand, the mandatory exercises are designed to ensure a good level of the data mining and machine learning fundamentals, some of their mathematical frameworks and to develop intuition about the subjects. It is recommended therefore that every group complete all the weekly exercise lists. The work can be divided among the group members.

General Concepts For the dataset provided, you must apply at least two pre-processing methods and one supervised learning method (*regression* or *classification*). For example, you could use *normalization*, *missing value replacement*, and *linear regression*. However, many other combinations are possible. You must implement all the code for these algorithms (in Python Scripts or Jupyter Notebooks). Additionally, your source code will only be accepted if it includes reasonable and clear comments.

Deadline The deadline for submitting Assignment #01 is Friday, 25 October 2026, at 23:59. Please submit your solution via the **Assignment #01** resource available on learnIT.

Teamwork You must complete this project in groups. We recommend you completing this project in groups of 3-4 students, but groups of 2 are allowed (groups with 1 member will automatically fail). Therefore, you will work together to formulate the questions, implement

the code, conduct the experiments, write the report, and implement the mandatory exercises. By default, all group members will receive the same grade (*passed* or *failed*). To get started, please inform your group members as soon as possible and record your group's composition in the Group Self-Selection resource on learnIT before the project submission deadline. Since group work can sometimes be uneven, you must include a one-page document titled “*Work Allocation*,” which details who was responsible for each part of the coding and report.

Submission Information Your submission must include (i) *source code*, (ii) *mandatory exercises implementations* and (iii) *project report*. Please compress these three parts into a **single zip file** named `groupXY_solution.zip`, where `XY` represents your group's ID (e.g., `group01_solution.zip`). Only one group member must submit the solution on learnIT.

Source Code The source code must be implemented in *Python* or *Jupyter Notebooks* and include all the necessary code and data to answer your defined questions. The code must be well-commented and easy to read. Minor technical errors will not be a cause for failure, at least as long as they are acknowledged within the report.

Mandatory Exercises and Project Report The mandatory exercises to be implemented and submitted are marked in each weekly exercise as (**Assignment #01**). The group must implement all mandatory exercises and deliver them as `.py` or `.ipynb` files. For the data mining and machine learning project, you must submit a two-page report as a `.pdf` file. In the project report, each page should contain up to 2,400 units (including spaces and notes). You must describe the questions you aimed to answer, the methods you used, the results you obtained, any problems you encountered (such as issues with the dataset), and any other relevant observations. Include numerical results and plots. If your report exceeds two pages due to layout or plots, please include the total unit count (including spaces and notes) in the report. It is allowed to add an Appendix with extra material, as long as the report itself is self-contained, with some references to the Appendix.

Grading We will grade your assignment based on a *pass/fail* basis. The final grade is dependent of the acceptance of the TAs following quality criteria, and the changes for the reassignment are mandatory. Completing the assignment will not impact your final grade. If you fail the assignment or miss the submission deadline, you can resubmit it after the group project deadline. However, the resubmission may require you to implement additional algorithms, and the new deadline may interfere with your exam preparation. Please note that passing the resubmission is mandatory to be eligible for the final oral examination.

Plagiarism and Code Reuse Plagiarism will not be tolerated and result in the plagiarist failing the assignment. On the other hand, the assignment is a natural continuation of the work done during the labs. Therefore, you can reuse the code you wrote during the labs and even some data set analysis you performed. If you completed all the labs, you will have most of the work on the assignment. Therefore, use the lab sessions effectively, and do not be afraid of asking the TAs and teacher any questions related to homework, coding and assignments, plus the ones beyond the scope of the course!

Use of Generative Artificial Intelligence The use of generative artificial intelligence is permitted during this assignment, provided that students comply with the official ITU guidelines on generative AI.